ROGERS CORPORATION

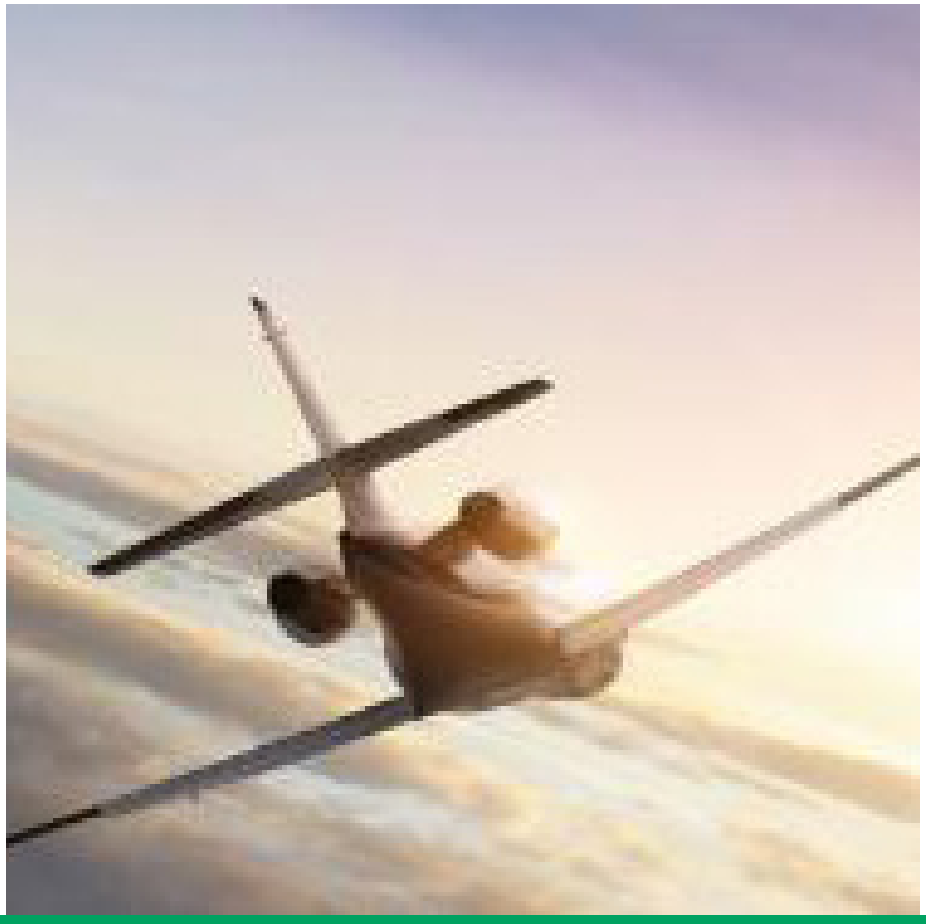
\\\ Datasheet

CLTE™ and CLTE-XT™ Circuit Materials
High Frequency Laminates

CLTE™ laminates have proven excellent dimensional stability and low planar CTE, providing consistent performance for embedded resistors: among the lowest variance available for PTFE-based laminates.

CLTE laminates have a long history of use with Resistor Foil and are available with a full range of other cladding types (including electrodeposited, reverse treated copper, rolled copper foil and more).

CLTE laminates' tried and tested performance continues to make them a top choice for a wide range of ground-based and airborne communications and radar systems. .



\\\ Features and Benefits:

Loss Tangent of .0012 at 10 GHz

- Reduced circuit losses without sacrificing dimensional stability

Low Z-axis CTE of 20 ppm /°C

- High plated through hole reliability

Dielectric constant stability with temperature change

- Reduced stress attachment to ceramic active devices

Available with heavy metal backing (aluminum, brass and copper)

- Reliably designed with embedded resistor networks

\\\ Typical Applications:

- Advanced Driver Assistance Systems (ADAS)
- Patch Antennas
- Phased Array Antennas
- Power Amplifiers

\\\ Standard Properties Table

Properties	Typical Values[1]		Units	Test Conditions		Unit
	CLTE	CLTE-XT™				
Electrical Properties						
Dielectric Constant (process)	2.98	See Table Below	-	23˚C @ 50% RH	10 GHz	IPC TM-650 2.5.5.5
Dielectric Constant (design)	2.98	2.93	-	C-24/23/50	10 GHz	Microstrip Differential Phase Length
Dissapation Factor	0.0021	0.0010	-	23˚C @ 50% RH	10 GHz	IPC TM-650 2.5.5.5
Thermal Coefficient of Dielectric Constant	6	-8	ppm/˚C	-50 to 150˚C	10 GHz	IPC TM-650 2.5.5.5
Volume Resistivity	1.4 X 10^9	4.25 X 10^8	Mohm-cm	C-96/35/90	-	IPC TM-650 2.5.17.1
Surface Resistivity	1.30 X 10^6	2.49 X 10^8	Mohm	C-96/35/90	-	IPC TM-650 2.5.17.1
Electrical Strength (dielectric strength)	1100	1000	V/mil	-	-	IPC TM-650 2.5.6.2
Dielectric Breakdown	64	58	kV	D-48/50	X/Y Direction	IPC TM-650 2.5.6
PIM	-	-	dBc	-	50 ohm 0.060"	43dBm 1900 MHz
Thermal Properties						
Decomposition Temperature (Td)	538	539	˚C	2hrs @ 105˚C	5% Weight Loss	IPC TM-650 2.3.40
Coefficient of Thermal Expansion - x	9.9	12.7	ppm/˚C		-55˚C to 288˚C	IPC TM-650 2.4.41
Coefficient of Thermal Expansion - y	9.4	13.7	ppm/˚C		-55˚C to 288˚C	IPC TM-650 2.4.41
Coefficient of Thermal Expansion - z	57.9	40.8	ppm/˚C		-55˚C to 288˚C	IPC TM-650 2.4.41
Thermal Conductivity	0.5	0.56	W/(m·K)		z direction	ASTM D5470
Time to Delamination	>60	>60	minutes	as-received	288˚C	IPC TM-650 2.4.24.1
Mechanical Properties						
Copper Peel Strength after Thermal Stress	1.2 (7)	1.7 (9)	N/mm (lbs/in)	10s @288˚C	35 µm foil	IPC TM-650 2.4.8
Flexural Strength (MD, CMD)	92.4, 86.9 (13.4, 12.6)	40.7, 40.0 (5.9, 5.8)	MPa (ksi)	25˚C ± 3˚C	-	ASTM D790
Tensile Strength (MD, CMD)	73.8, 71.0 (10.7, 10.3)	29.0, 25.5 (4.2, 3.7)	MPa (ksi)	23C/50RH	-	ASTM D638
Flex Modulus (MD, CMD)	8122, 7984 (1178, 1158)	3247, 3261 (471, 473)	MPa (ksi)	25˚C ± 3˚C	-	ASTM D790
Dimensional Stability (MD, CMD)	-0.07, -0.02	-0.37, -0.67	mm/m	4 hr at 105˚C	-	IPC-TM-650 2.4.39a
Physical Properties						
Flammability	V-0	V-0	-	-	C48/23/50 & C168/70	UL 94
Moisture Absorption	0.04	0.02	%	E1/105+D24/23	-	IPC TM-650 2.6.2.1
Density	2.31	2.17	g/cm^3	C-24/23/50	-	ASTM D792
Specifc Heat Capacity	0.60	0.61	J/g˚K	2 hours at 105˚C	-	ASTM E2716
NASA Outgassing — Total Mass Lost	0.02	0.02	%	-		ASTM E595
NASA Outgassing — Collected Volatiles	0.00	0.00	%			

[1] Typical values are a representation of an average value for the population of the property. For specification values contact Rogers Corp.

\\\ Property Charts

Chart 1

Microstrip Differential Phase Length Method , Dk vs Frequency

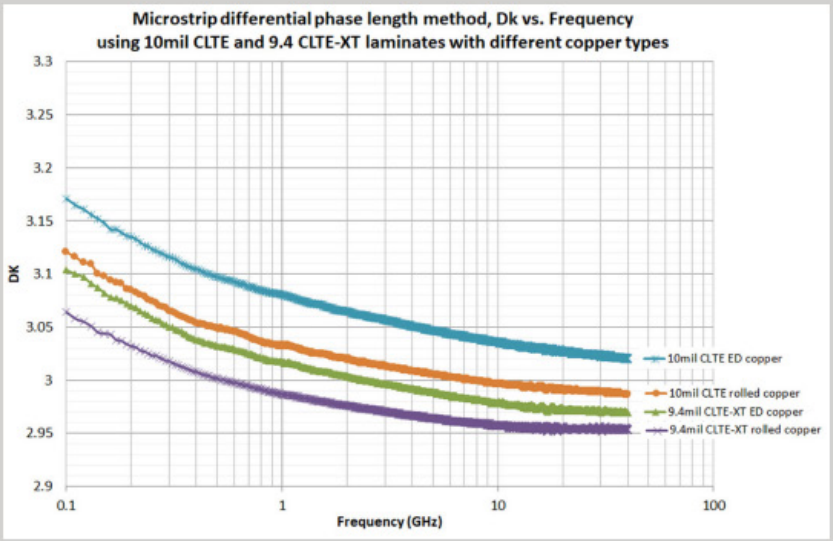
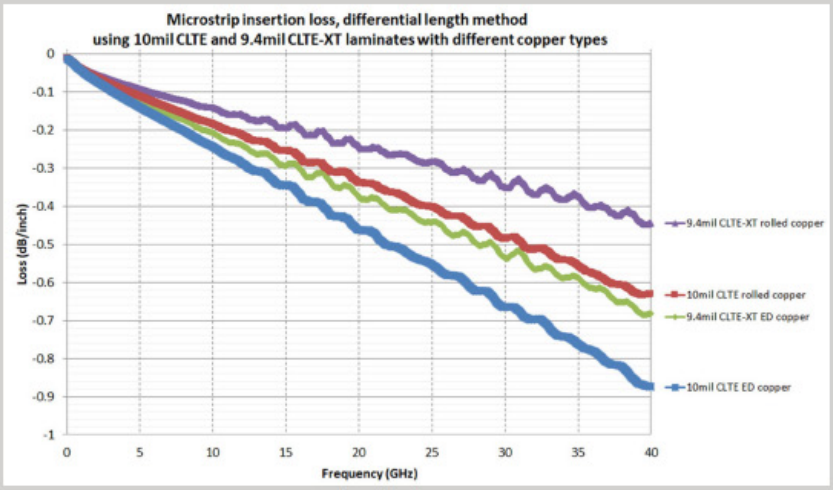


Chart 2

Microstrip Insertion Loss Differential Length Method With Different Copper Types



\\\ CLTE-XT Dielectric Constant Table

Grade	Panel Thickness	Process Dk	Tolerance
Dk by Thickness			
CLTE-XT	0.0051" (0.135 mm)	2.79	± 0.03
	0.0094" (0.254 mm)	2.89	± 0.03
	0.020" (0.508 mm)	2.92	± 0.03
	0.030" (0.762 mm)	2.94	± 0.03

\\\ Standard Offerings

Standard Thicknesses		Standard Panel Sizes	Standard Claddings
CLTE: 0.0053" (0.135 mm) ± 0.0005" 0.010" (0.254 mm) ± 0.0010" 0.020" (0.508 mm) ± 0.0020" 0.030" (0.762 mm) ± 0.0020"	**CLTE-XT:** 0.0051" (0.130 mm) ± 0.0005" 0.0094" (0.239 mm) ± 0.0007" 0.020" (0.508 mm) ± 0.0010" 0.030" (0.762 mm) ± 0.0010"	18" X 12" (457 X 305mm) 18" X 24" (457 X 610mm)	Electrodeposited Copper Foil 1/2 oz. (18µm) 1 oz. (35µm) Reverse Treated Electrodeposited Copper Foil 1/2 oz. (18µm) 1 oz. (35µm)

*Contact Customer Service or Sales Engineering to inquire about other available product configurations including additional thicknesses, panel sizes and claddings.

\\\ 100 S. Roosevelt Avenue \\\ Chandler, AZ 85226 \\\ Tel: 480-961-1382 \\\ Fax: 480-961-4533 \\\ www.rogerscorp.com
\\\ IPC Slash Sheet # 4103B/006 \\\ UL File #

The information in this preliminary data sheet is intended to assist you in designing with Rogers' circuit materials. It is not intended to and does not create any warranties express or implied, including any warranty of merchantability or fitness for a particular purpose or that the results shown on this data sheet will be achieved by a user for a particular purpose. The user should determine the suitability of Rogers' circuit materials for each application. These commodities, technology and software are exported from the United States in accordance with the Export Administration regulations. Diversion contrary to U.S. law prohibited.

The Rogers' logo, Helping power, protect, connect our world, CLTE and CLTE-XT are trademarks of Rogers Corporation or one of its subsidiaries.
©2023 Rogers Corporation, Printed in U.S.A., All rights reserved.

Issued 1665 101623 **Publication #92-201**

Helping power, protect, connect our world™

ROGERS CORPORATION